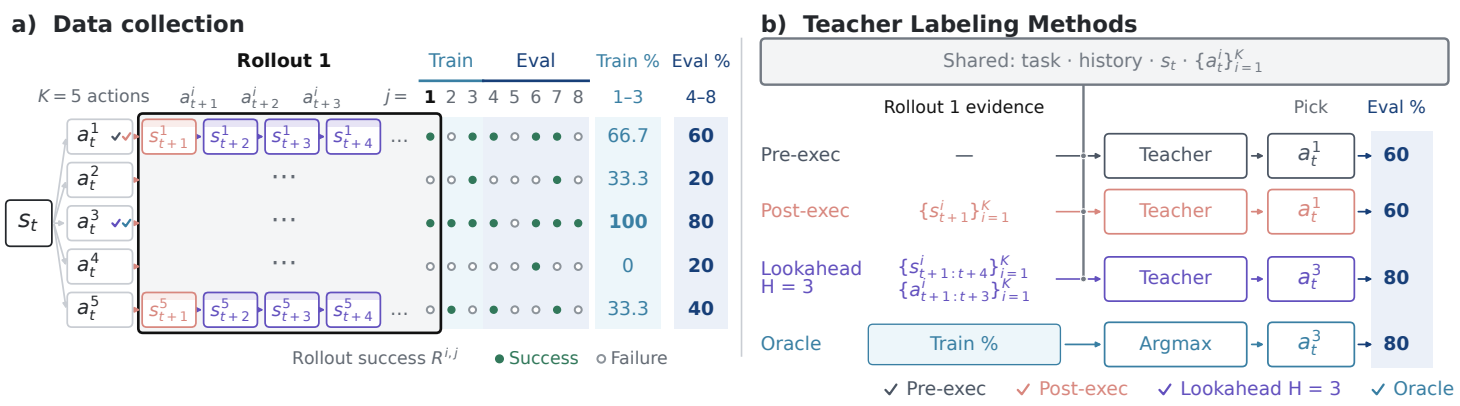
a) Data collection
Rollout 1
Train
Eval
Train %
Eval %
K = 5 actions
a^i_{t+1} a^i_{t+2} a^i_{t+3}
j = 1 2 3 4 5 6 7 8
1–3
4–8
s_t
a^1_t
a^2_t
a^3_t
a^4_t
a^5_t
s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4}
s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4}
66.7 60
33.3 20
100 80
0 20
33.3 40
Rollout success $R^{i,j}$
Success
Failure
b) Teacher Labeling Methods
Shared: task · history · s_t · $\{a^i_t\}_{i=1}^K$
Rollout 1 evidence
Pick
Eval %
Pre-exec
—
Post-exec
$\{s^i_{t+1}\}_{i=1}^K$
Lookahead H = 3
$\{s^i_{t+1:t+4}\}_{i=1}^K$
$\{a^i_{t+1:t+3}\}_{i=1}^K$
Oracle
Train %
Teacher
Argmax
a^1_t
a^3_t
60
60
80
80
✓ Pre-exec
✓ Post-exec
✓ Lookahead H = 3
✓ Oracle